

The Weight of Nothing: Missing Electrodes, Empty Baselines, and the Edge of Stability in Spiking Reservoir Encoding of Affective EEG

Andrew A. Lane, Wendy Tang, and Brady D. Nelson

Abstract—How representations of affective event-related potentials (ERPs) degrade under perturbation depends on what an evaluation lets stand for nothing: the number replacing a missing electrode, the origin of each feature coordinate, what the pre-stimulus baseline holds after normalization, and the chance floor. We test them on a fixed leaky integrate-and-fire spiking reservoir (six spike-count bins), band-power, ERP-window, and EEGNet, under subject-grouped validation on the SHAPE cohort (211 subjects) and a public IAPS cohort (228 subjects). Moving the coordinate origin leaves clean predictions unchanged, up to solver tolerance, while channel-dropout accuracy falls when the fill moves far from the training mean. The zero-versus-mean fill choice rewrites predictions in the order of feature non-centeredness (34.5% of band-power, 32.9% of ERP-window, 3.6% of reservoir labels) but costs accuracy only as far as the above-chance margin allows. Per-epoch z-scoring writes the post-stimulus mean into the pre-stimulus baseline ($r = -0.91$), enough for pre-stimulus samples alone to classify affect above chance (permutation $p = 0.025$), and the conference reservoir window gains $+0.092$ BA when moved after onset. Across ten reservoir draws the measured edge of stability lies at $\rho^* = 0.76 \pm 0.04$, below $\rho = 1$, but the firing threshold moves it from 0.35 to at least 1.57, and past the edge a silent electrode lies 5 to 6 standard deviations from typical reservoir activity. On the long external epoch, spline repair of removed electrodes erases the encoder ranking that zero traces produce. We recommend reporting fills with feature centeredness and above-chance margin, normalization and window placement, and operating points across draws.

Index Terms—EEG, event-related potentials, spiking neural networks, reservoir computing, missing data, normalization.

I. INTRODUCTION

Electroencephalographic event-related potentials (ERPs) measure evoked neural dynamics at millisecond resolution, but the recorded waveform is a noisy, low-dimensional, and subject-dependent projection of an unobserved neural process [3], [4]. A representation of affective ERPs can separate conditions under one protocol and keep little usable structure after amplitude noise, temporal misalignment, or electrode

loss, and endpoint accuracy alone does not say which. Spiking reservoirs make the gap sharper: fixed recurrent networks give nonlinear temporal expansions [5], [6], yet EEG studies usually score them as classifiers [7], [8] rather than as measurement operators whose invariances can be tested.

Every such test has to decide what *nothing* means. An absent electrode must be represented by some number. A feature coordinate has an origin, and a zero-filled block sits at that origin whether or not it means anything there. The interval before stimulus onset contains no stimulus-locked signal, yet it is part of every epoch the encoder reads. A score has to be compared with the floor set by no information at all. And a reservoir’s recurrence can be switched off ($\rho = 0$) to ask what the dynamics add. These choices are rarely stated, and it has not been tested how far each one moves a measured result or whether its effect transfers between cohorts.

Our conference study [2] measured two of them on one cohort. On SHAPE (211 subjects), raw retention rewarded the encoder closest to chance, and zero-filling a dropped channel before standardization, which is not invariant to re-centering the features (Proposition 1), halved the measured lead of ERP-window features over a near-centered spiking-reservoir code at 30% dropout ($+0.049$ versus $+0.092$ to $+0.107$ BA), and on DEAP the fill rule alone moved band-power BA by up to 0.11. The reservoir, a fixed leaky integrate-and-fire (LIF) network read out through six spike-count bins (BSC_6), sat just past its measured edge of stability ($\rho^* \approx 0.82$, below the echo-state heuristic $\rho = 1$), and removing its recurrence cost 0.049 BA. That study left five questions open. It used one cohort, one reservoir draw, and one EEGNet recipe. It argued Proposition 1 algebraically but never moved the origin to watch the score follow. And its own camera-ready code check found that its reservoir window, $t \in [10, 70)$ samples, spans about -160 to $+73$ ms around onset, so most of what the reservoir encoded came before the stimulus.

This paper answers those questions on SHAPE itself, rerun from its data with new experiments, and on a second, public cohort. The contributions are:

- 1) A rerun of SHAPE that reproduces the conference’s main analyses exactly and adds analysis windows measured from true onset, and an external, paradigm-matched test on an independent IAPS cohort (228 subjects, different laboratory and montage) at the conference-matched epoch and at a long epoch pre-specified from the dataset’s published effect latencies (Secs. IV, V-A).

A. A. Lane and W. Tang are with the Department of Electrical and Computer Engineering, Stony Brook University, Stony Brook, NY 11794, USA.

B. D. Nelson is with the Department of Psychology, Stony Brook University, Stony Brook, NY 11794, USA.

A preliminary version of this work was accepted at the 2026 IEEE International Conference on Bioinformatics and Biomedicine (BIBM) [2]. This version reruns SHAPE with new experiments and with analysis windows measured from stimulus onset, and adds an independent public cohort, a direct test of Proposition 1, Proposition 2 and the pre-stimulus experiments, the operating point across ten reservoir draws, spherical-spline repair and recorded trial-count noise, and EEGNet recipe variation.

- 2) A direct experimental test of Proposition 1 on both cohorts: shifting the coordinate origin leaves clean predictions unchanged up to solver tolerance, while the dropout score falls when the fill moves far from the training mean (Sec. V-C).
- 3) Proposition 2, which shows that per-epoch z-scoring writes a function of the post-stimulus signal into the pre-stimulus interval, and the test it implies: on SHAPE the conference reservoir window read that offset rather than evoked dynamics, and the same reservoir placed after onset measures the evoked signal (Secs. III-G, V-D).
- 4) The operating point across ten reservoir draws on both cohorts, and across thresholds and leaks: where the measured edge of stability lies and whether accuracy follows it, judged by a decision rule fixed before the SHAPE sweep ran (Sec. V-E).
- 5) Two physically grounded perturbations, spherical-spline repair of removed electrodes and re-averaging from fewer recorded trials, together with EEGNet with and without augmentation and across recipes on the second cohort (Sec. V-G).

The affective categories define the task. The object of study is what each representation does with nothing: with a missing electrode, a shifted origin, an empty baseline, and a chance floor.

II. RELATED WORK

ERP representations of affect. Emotional pictures enlarge the late positive potential relative to neutral ones from roughly 400 ms after onset [9]–[11], and a recent cohort of 224 participants adds an ordered valence effect between 1500 and 2500 ms [12]. ERP/EEG decoders commonly use window amplitudes, Hjorth [13] and spectral descriptors [14], and linear models [4], [15], which are interpretable through these known components but can miss nonlinear temporal structure. Deep networks such as EEGNet [16] are now common [8], [17], and much of this literature reports endpoint accuracy over preprocessing and acquisition choices, which can change measured performance materially [18]–[20].

Reservoirs and the edge of stability. Reservoir computing maps an input through a fixed high-dimensional recurrent system to a state trajectory read out by a simple classifier [5], [6], [21], [22], and spiking liquid-state machines add event timing and count structure [23], [24]. Computation is often reported to peak near the transition between ordered and irregular dynamics, located by how far a small state perturbation spreads [25]–[27]. The spectral radius $\rho = 1$ is a heuristic for rate reservoirs, and it is neither necessary nor sufficient for the echo-state property of a driven system [28], so for a thresholded spiking reservoir the transition need not lie at $\rho = 1$. Neuromorphic reservoir features for affective EEG were introduced in earlier ARSPI-Net work [29], [30], and this paper extends our conference study [2].

Missing electrodes. Dynamic spatial filtering handles missing channels with a learned attention front end [31], and channel-dropout or noise augmentation hardens trained models [32]. A recent EEG foundation-model benchmark finds

that apparent channel-dropout fragility depends on whether dropped channels are removed or zero-padded [33], and an evaluation of four EEG decoding models finds channel permutation far more damaging than channel removal [1]. Both observe that the representation of an absent or misplaced channel moves the measured result. Proposition 1 below gives the mechanism for standardized features and the condition under which a fill is harmless. Fills range from constants and spherical-spline interpolation [34], through learned imputation [31], to generative synthesis of whole missing modalities [35], [36]. The non-learned fills studied here sit at the simple end of that spectrum.

Evaluation choices that create or remove effects. Filtering can move late ERP effects to earlier latencies and create artifactual early effects [37]–[39], and leakage between preprocessing and evaluation inflates reported accuracy across machine-learning science [40], [41]. Decoding accuracy near a theoretical chance level needs empirical null distributions [42], and cross-validated accuracy on a few hundred observations carries wide error bars [43]. Per-epoch normalization, the fill convention for an absent electrode, and the origin of a feature coordinate belong to the same family of choices. We give each a proposition or a direct test.

III. METHODS

A. ERP Observation Model

Let $X_s^{(c)} \in \mathbb{R}^{C \times T}$ denote the trial-averaged ERP observation of subject s in condition c . We treat it as a noisy readout of an unobserved neural state trajectory,

$$\xi_{t+1} = f_\theta(\xi_t, u_t) + \eta_t, \quad x_t = H\xi_t + \epsilon_t, \quad (1)$$

where H is the scalp measurement operator and ϵ_t collects measurement noise, residual single-trial variability, and preprocessing. Every perturbation below acts on a term of this model: electrode removal zeroes rows of H , additive noise and re-averaging from fewer trials enlarge ϵ_t , and alignment jitter shifts the sampling index of x_t . Normalization acts on x_t before any encoder sees it, which is why it is treated as part of the measurement (Sec. III-G).

B. Spiking Reservoir Encoding

Each electrode trace drives a fixed LIF reservoir of 256 units with input weights W_{in} , recurrent weights W , leak $\beta = 0.05$, and threshold $\vartheta = 0.5$:

$$v_t = [(1 - \beta)(1 - z_{t-1}) \odot v_{t-1} + Wz_{t-1} + W_{\text{in}}x_t]_+, \quad (2)$$

$$z_t = \mathbf{1}[v_t \geq \vartheta], \quad (3)$$

where v_t is the membrane potential, z_t the binary spike vector, and $[\cdot]_+$ rectification at zero. The factor $(1 - z_{t-1})$ resets every unit that spiked at the previous step. W_{in} and W are drawn once from a Xavier-uniform law, and W is rescaled to spectral radius ρ . Unless stated otherwise the draw is seed 42 and $\rho = 0.9$, the conference operating point, and only the readout is trained. Figure 1 shows the reservoir’s activity and the conference operating-point results on SHAPE.

Activity is summarized by six equal temporal bins over an analysis window. For unit j and bin k , the binned spike-count code is $b_{j,k} = \sum_{t \in \mathcal{B}_k} z_j(t)$. Each electrode's 1536-dimensional spike-bin vector is projected onto a single 64-component PCA basis shared across electrodes and fitted on the training subjects of each fold, and the projections are concatenated, giving $E \in \mathbb{R}^{64C}$ (2176 for SHAPE, 2048 for the external cohort):

$$E = \phi_{\text{PCA}}([b_1, \dots, b_6]) = \Phi(X). \quad (4)$$

A companion study compares this data-fitted projection with a fixed random projection and examines how subject covariance shapes affective EEG embeddings [44]. Here we keep the conference's train-only PCA so that every result is comparable with the conference version. The conference window is $t \in [10, 70]$ in samples. With onset at sample 51 in both cohorts it spans about -160 to $+73$ ms around onset, so most of it precedes the stimulus. Section III-G examines what that window can read, and Sec. V-D adds windows wholly before and wholly after onset. We also rerun the conference analyses with windows measured from true onset, both fixed before any of their results was computed: the window the conference text described, $+39$ to $+273$ ms ($t \in [61, 121]$, the *onset* window), and the whole post-onset interval, 0 to $+793$ ms ($t \in [51, 255]$, six bins of 34 samples, the *post* window). In both reruns the ERP-window encoder uses its three windows measured from onset (80–200, 250–450, and 450–800 ms), as the conference labels described them.

A silent electrode is a fixed point of the reservoir: with $x_t = 0$ from a zero initial state, v_t stays at zero and no unit spikes, so the electrode's spike-bin block is exactly zero. After the PCA step that block maps to $-P\bar{b}$, the negative of the projected training mean, not to the origin of the embedding.

C. Operating Point and Order Parameter

To test whether the recurrent dynamics matter, the same draw is rescaled to $\rho \in \{0, 0.3, 0.6, 0.9, 1.2, 1.5\}$ with every other parameter fixed ($\rho = 0$ leaves a feedforward LIF layer). A label-free order parameter locates the dynamical transition by damage spreading [25]: two copies of the reservoir receive the same electrode drive, one unit's spike is flipped at $t = 10$, and the normalized Hamming distance between the copies' spike vectors is averaged over steps 21 to 69 and over 400 sampled drives and flipped units. The transition ρ^* is the ρ at which this damage, linearly interpolated on the grid $\rho = 0, 0.1, \dots, 2$, reaches half its maximum. The conference measured ρ^* and accuracy for one draw on SHAPE. Here both are repeated for ten independent draws (seeds 42–51) on SHAPE with the conference definition, accuracy clean and with 30% of the electrodes silent, and the accuracy sweep is repeated with the onset and post windows. On the external cohort the order parameter uses ten draws on the drives of the conference-matched epoch, and the accuracy sweep five draws (seeds 42–46) on the long epoch (Sec. IV), where the reservoir is above chance. For draw 42 the transition is also measured at thresholds $\vartheta \in \{0.25, 0.5, 1.0\}$ and leaks $\beta \in \{0.02, 0.05, 0.1, 0.2\}$ on both cohorts' drives. Before

the SHAPE sweep ran we fixed the decision rule that the conference review proposed: the recurrence effect stands if the deficit at $\rho = 0$ relative to $\rho = 0.9$ keeps its sign in at least nine of ten draws and the interval of the draw-mean difference, computed on the same subject resamples for every draw, excludes zero. Otherwise only a broad optimum is claimed.

D. Baselines

Band-power uses Welch power in five bands (1–4, 4–8, 8–13, 13–30, 30–45 Hz) per channel (5C dimensions). ERP-window uses mean amplitude in three windows, about -120 – 0 ms, 50 – 250 ms, and 250 – 600 ms relative to onset, per channel (3C dimensions). All three fixed encoders feed a balanced L2-regularized logistic readout. As a trained baseline, EEGNet-8,2 [16] is trained per fold on the same per-epoch z-scored $C \times T$ signal that the fixed encoders read (Adam, learning rate 10^{-3} , 80 epochs, batch 64, dropout 0.25) with train-only per-channel standardization, with and without augmentation by random channel dropout and additive noise. On the external cohort the recipe is also varied (40 or 160 epochs, dropout 0.5, conference-matched epoch) to probe the effect of a single recipe. For cost, the recipe variants and EEGNet on the long epoch use one partition (seed 42, five folds) instead of five, with intervals from the same subject bootstrap.

E. Perturbations

The feature-coordinate experiment suppresses a dropped channel's feature block after extraction and refills it by one of four rules: zero, train mean, train-fitted k -nearest-neighbor imputation ($k=10$), and the mean of the retained channels (spatial). It isolates the fill convention and applies to the fixed encoders.

The signal-level experiment corrupts the test subjects' signal before any feature extraction, identically for every encoder including EEGNet, with the same draws in each fold, while training uses the clean signal: removal of 10, 30, or 50% of the electrodes (a removed electrode is a zero trace), additive white noise at 5 dB SNR per channel, and a uniform alignment shift of up to ± 50 ms. Two signal-level conditions are new. *Spline repair* replaces each removed electrode by spherical-spline interpolation from the retained electrodes [34] (order 4, seven Legendre terms) on the baseline-corrected microvolt signal before normalization, a physically grounded alternative to the zero trace. *Recorded observation noise* re-averages every test observation from 1, 4, or 16 of its own single trials instead of all of them, so the added variance is the recorded trial-to-trial variability rather than synthetic noise. It needs single trials and is run on the external cohort only.

F. Retention and the Coordinate Origin

Raw retention, the ratio of perturbed to clean BA, overstates the stability of representations whose clean BA is close to chance ($c = 1/3$), so we also report *above-chance retention*, $(\text{BA}_{\text{pert}} - c)/(\text{BA}_{\text{clean}} - c)$, and treat absolute BA and paired subject-level differences as primary.

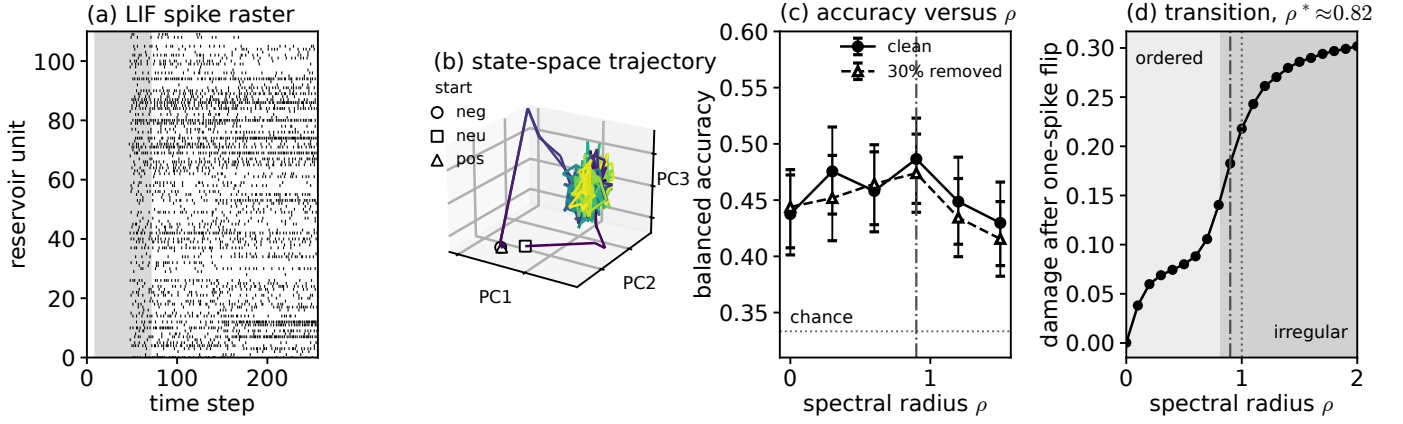


Fig. 1. The fixed LIF reservoir as a dynamical system (SHAPE, draw 42, conference results [2]). (a) Spike raster (units 0–109 of 256) for one condition. Shaded = the conference window $t \in [10, 70]$, about -160 to $+73$ ms around onset at step 51. (b) Population state-space trajectory (first three principal components of the membrane state) for the three conditions. (c) Subject-level BA (95% CI) versus spectral radius ρ , clean and with 30% of electrodes removed, with dash-dot = the pre-specified $\rho = 0.9$. (d) Damage-spreading order parameter versus ρ . Shading separates the ordered and irregular regimes at $\rho^* \approx 0.82$, and dotted = the echo-state heuristic $\rho = 1$.

Write the standardized readout input as $z = D_\sigma^{-1}(x - \mu)$ with $D_\sigma = \text{diag}(\sigma)$, and let S index the dropped coordinates. Zero-fill sets $x_S = 0$, so the dropped block enters as $z_S = -D_{\sigma,S}^{-1}\mu_S$, and train-mean fill sets $x_S = \mu_S$, so $z_S = 0$:

$$z_S^{\text{zero}} - z_S^{\text{mean}} = -D_{\sigma,S}^{-1}\mu_S. \quad (5)$$

Proposition 1 (origin dependence of zero-fill). *Let the clean readout depend on x only through z , so it is invariant to a constant offset $x \mapsto x - b$ (train-only standardization guarantees this). Then the clean prediction is unchanged by the re-centering, while the zero-fill dropout input shifts by $\Delta z_S = D_{\sigma,S}^{-1}b_S$. Hence the measured zero-fill dropout accuracy depends on the coordinate origin, and the zero- and mean-fill inputs coincide if and only if $\mu_S = 0$.*

Proof. Standardization subtracts the training mean, so b cancels from the clean input. A zero-filled coordinate maps to $z_i = -(\mu_i - b_i)/\sigma_i$ instead of $-\mu_i/\sigma_i$, a shift of $b_i/\sigma_i \neq 0$ for any $b_i \neq 0$. $\square$

For a linear readout with class weights W , Eq. (5) moves the class logits by $\Delta \ell = -W_S D_{\sigma,S}^{-1}\mu_S$ between the two fills, so the fills can disagree only as far as the dropped features are non-centered. Proposition 1 has a direct experimental test, which the conference study did not run: fill the dropped block with the raw value $x_i = \mu_i + \kappa\sigma_i$, which places it at standardized position $z_i = \kappa$ for every encoder, and sweep κ . Clean predictions must not change with any offset b , while dropout accuracy can depend on κ , with train-mean fill at $\kappa = 0$. An encoder’s own zero places each dropped feature at its own $z_i = -\mu_i/\sigma_i$, at a mean absolute standardized distance $|\mu|/\sigma$ from the training mean, so it is not a single point on the κ axis. We refit the readout on features shifted by $b = \mu + 2\sigma$ to check the first prediction and sweep $\kappa \in [-3, 3]$ to test the second.

G. What the Pre-Stimulus Interval Can Contain

Both cohorts are baseline-corrected over the 200 ms before onset, so the pre-stimulus mean of every trial-averaged trace

is zero at the recording rate and close to zero after decimation, and both are then z-scored per channel over the whole epoch, as in the conference pipeline. Let P be the pre-stimulus samples and Q the rest, with $|P| + |Q| = T$, and let m_Q be the post-stimulus mean and s the epoch standard deviation of one channel.

Proposition 2 (baseline transfer under per-epoch normalization). *For a trace with zero pre-stimulus mean normalized by its own epoch mean and standard deviation, the mean of the normalized pre-stimulus segment is $-|Q|/T m_Q/s$. The pre-stimulus segment is therefore a deterministic function of the post-stimulus signal, and any encoder that reads only P has access to m_Q/s .*

Proof. The epoch mean is $(|P| \cdot 0 + |Q| m_Q)/T$. Subtracting it and dividing by s maps the zero pre-stimulus mean to $-|Q|m_Q/(Ts)$. $\square$

With onset at sample 51 of 256, $|Q|/T = 0.80$. An affective modulation of the late positive potential changes m_Q , so under per-epoch z-scoring a classifier can separate conditions from an interval in which no stimulus has yet been shown. A fixed per-channel scale (one standard deviation per channel, pooled over all observations without labels and applied identically to every epoch) leaves the pre-stimulus mean where baseline correction put it and removes this route. Because the scale is shared by all epochs, it cannot carry information from one part of an epoch into another, although it is transductive with respect to subject inclusion. We compare the two normalizations for windows wholly before onset, the conference window, and windows wholly after onset (Sec. V-D). Proposition 2 also says what a reservoir window before onset can encode. Its drive there is nearly constant at the transferred offset, and a fixed reservoir driven by a nearly constant input encodes that constant in its spike counts. We test this directly. For every trace we predict the post-stimulus mean m_Q/s from the spike-bin code of a window wholly before onset ($t \in [3, 51]$) and of the conference window, with a ridge regression (penalty 1) fitted on four-fifths of the subjects and scored by out-of-fold R^2 , under per-epoch z-scoring and under the fixed scale.

We also report the per-channel correlation between the total spike count in the conference window and m_Q/s , and with $|m_Q/s|$, because the rectified membrane and input weights of both signs let an offset of either sign raise the count.

IV. DATA AND PROTOCOL

A. SHAPE Cohort

The primary cohort is the trial-averaged affective ERP data of the Stress, Health, and the Psychophysiology of Emotion (SHAPE) project [45], collected by the Laboratory for Clinical Affective Neuroscience at Stony Brook University: 211 subjects after exclusion of one anomalous record, each contributing negative, neutral, and pleasant observations (633 in total). Each epoch (34 scalp channels at 1024 Hz, 1.2 s, baseline-corrected over the 200 ms pre-stimulus interval) is analyzed as every fourth of its first 1024 samples: 256 samples at 256 Hz spanning -200 to $+800$ ms, onset at sample 51, z-scored per channel within each epoch (Fig. 2). For this version every new experiment was also run on SHAPE. We checked that the conference input equals the z-scored, every-fourth-sample microvolt epoch to the last bit for all 633 observations, and the journal code reproduces exactly the conference values it repeats: clean BA and intervals of the three fixed encoders, all four fills at 30% dropout, and the draw-42 accuracy sweep and order parameter (21 values, largest difference 0, reservoir BA 0.487, CI [0.447, 0.523]). The pooled-basis sensitivity check differs from the conference value (0.485) by one observation of 633 (0.487), because the conference used an embedding regenerated by an earlier script with a different numerical path. The microvolt epochs give the fixed-scale normalization of Sec. III-G. The channel order of the SHAPE export is not documented, so no montage-dependent analysis (spline repair) runs on SHAPE, and only subject averages exist, so recorded trial-count noise is measured on the external cohort alone. SHAPE data are held in a restricted environment, and only aggregate results leave it.

B. External Cohort

The external cohort is a public IAPS passive-viewing dataset [12], [46] from a different laboratory and montage: 32 scalp electrodes (BioSemi layout) re-referenced to the mastoids and exported at 512 Hz after the dataset authors' topographic interpolation of bad channels, filtering, ICA ocular correction, and artifact rejection. Of 239 participants, the 228 with a complete negative, neutral, and pleasant panel in the preprocessed release are used (684 observations, median 44 trials per observation, range 20 to 54, with high- and low-arousal pictures of one valence pooled). Each observation is built to the SHAPE layout: trial average, baseline correction over -200 to 0 ms, every second sample from -200 ms, giving 256 samples at 256 Hz with onset at sample 51, and z-scoring per channel within the epoch. This conference-matched epoch is the primary external analysis. The grand averages reproduce the published effects (Fig. 3): emotional pictures exceed neutral ones by $2.12 \mu\text{V}$ at PO3 between 500 and 1300 ms, and negative pictures exceed pleasant ones by $1.32 \mu\text{V}$ between 1500 and 2500 ms. Most of both effects lies

after $+800$ ms, so we pre-specified, before computing any of its results, a secondary analysis on a long epoch from -200 to $+2496$ ms (691 samples, same sampling, baseline, and onset). In it the reservoir window covers 0 to $+2484$ ms after onset (six bins of 106 samples), the ERP-window encoder adds the two published windows (500–1300 and 1500–2500 ms), and band-power and EEGNet see the whole epoch.

C. DEAP Check

On DEAP [47] (32 subjects, band-power features, binary valence, within-subject 5-fold) the conference measured how much the fill rule alone moves channel-dropout accuracy. That check is reported unchanged (Fig. 8a).

D. Validation

All analyses share one fold structure: StratifiedGroupKFold (5 folds, seeds 42–46) with subjects as groups, so a subject's observations never span training and test. Standardization and the reservoir's PCA basis are fitted on the training subjects of each fold, the fixed encoders use a balanced L2 logistic readout, and EEGNet is trained on the same folds. The out-of-fold class probabilities of each observation are averaged over the five partitions, and 95% confidence intervals (CIs) come from 4000 resamples of the subjects with replacement, each keeping its three observations. Contrasts between encoders, fills, conditions, or operating points are paired differences under the same resampling, so no test treats the 25 overlapping folds as independent. Chance is $1/3$, and a within-subject label-permutation null (20 permutations, in which each subject keeps one observation per class) gives the empirical floor for every fixed encoder [42]. The pre-stimulus test of Sec. V-D uses 200 such permutations and reports $p = (1 + \#\{\text{null} \geq \text{observed}\}) / (1 + 200)$. The journal code reproduces the conference implementation bit for bit, on synthetic data (reservoir codes, damage, and bootstrap intervals) and on SHAPE.

V. RESULTS

A. Clean Classification on Two Cohorts

Table I reports clean three-class BA, macro-F1, macro one-vs-rest ROC-AUC, and the largest BA in the permutation null for every encoder and cohort. On SHAPE with the conference windows, ERP-window features are the strongest fixed encoder (BA 0.629), above band-power (0.494) and the reservoir (0.487), and EEGNet is more accurate still (0.731). With windows measured from true onset (Table II), the reservoir reaches 0.548 [0.515, 0.581] with the onset window ($+39$ to $+273$ ms) and with the post window (0 to $+793$ ms), and ERP-window 0.645 [0.611, 0.678]. The ERP-window lead over the reservoir, $+0.142$ (CI [$+0.098$, $+0.188$]) with the conference windows, is $+0.097$ (CI [$+0.054$, $+0.141$]) with the onset window and (CI) with the post window. The temporal bins now matter: BSC₆ exceeds the single-bin count BSC₁ by $+0.107$ (CI [$+0.065$, $+0.150$]) on the onset window and (CI) on the post window, whereas on the conference window the two were identical ($+0.001$). Section V-D shows what the conference window measured instead.

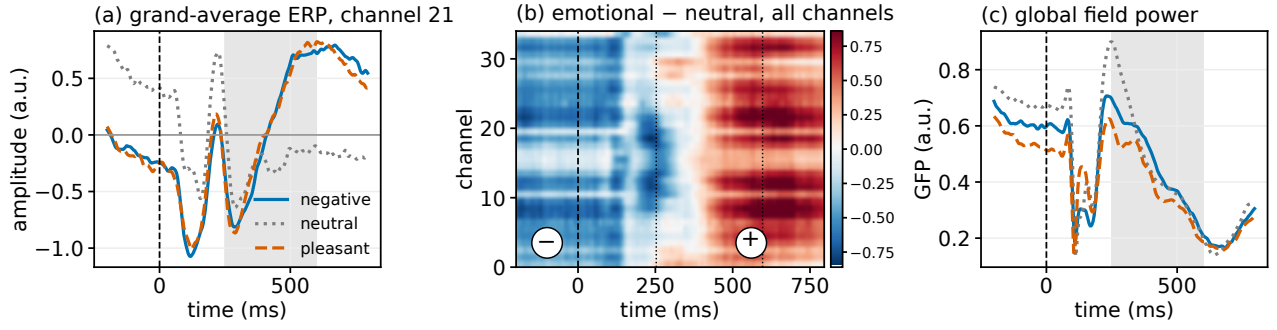


Fig. 2. Trial-averaged affective ERPs of SHAPE (grand averages over 211 subjects, z-scored per epoch, conference results [2]). (a) Waveforms per condition at the most affect-discriminative channel, with time 0 = onset (dashed), shaded = the late ERP-window feature (250–600 ms). (b) Emotional minus neutral across all 34 channels (–/+ mark the sign). (c) Global field power per condition.

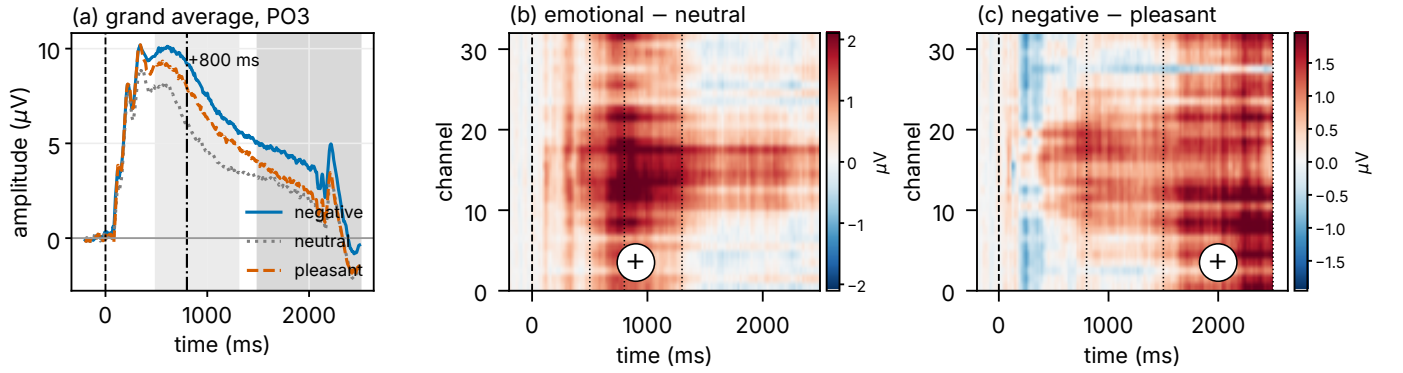


Fig. 3. Grand-average ERPs of the external cohort (228 subjects, baseline-corrected microvolts, long epoch). (a) Waveforms per valence at the channel with the largest emotional-minus-neutral difference between 500 and 1300 ms. Light and dark shading mark the published salience (500–1300 ms) and valence (1500–2500 ms) windows, and the dash-dot line the end of the conference-matched epoch (+800 ms). (b) Emotional minus neutral and (c) negative minus pleasant over all 32 channels, where + marks the sign of the effect in grayscale.

The external cohort carries much less condition information in the same layout. At the conference-matched epoch ERP-window features clearly exceed chance (BA 0.386, CI [0.358, 0.414], against a null maximum of 0.367). Band-power exceeds its null maximum only marginally (0.368 against 0.365, CI [0.339, 0.399]), and the reservoir is at chance (0.320, 95% CI [0.288, 0.354]). On the pre-specified long epoch every fixed encoder clears chance: ERP-window 0.427 [0.398, 0.455], reservoir 0.396 [0.361, 0.428], and band-power 0.389 [0.360, 0.418], and each exceeds its largest permutation BA (null maxima 0.361, 0.387, and 0.377), ERP-window by the widest margin. ERP-window leads the reservoir by +0.030 BA (CI [−0.012, +0.072]). ERP-window is the strongest fixed encoder on both cohorts, although band-power and the reservoir swap places on the long epoch, and the external above-chance margin is about a third of SHAPE’s (0.094 versus 0.295 for ERP-window), which limits the losses measurable on it. The grand averages reproduce the published effects (Fig. 3), so the pipeline recovers the cohort’s ERPs, but the effects that three-class decoding needs are small against the spread between subjects. At PO3, the channel with the largest salience effect (so these effect sizes are optimistic), the salience effect (2.12 μV , within-subject $d_z = 0.69$) is less than half the between-subject standard deviation of the neutral-condition amplitude in the same window (5.58 μV), and the negative-versus-pleasant effect is weak (1.32 μV , $d_z = 0.25$) and late.

The reservoir controls of the conference replicate in kind. Pooled (transductive) PCA instead of train-only PCA changes long-epoch BA by -0.006 ([−0.018, +0.004]), so the sensitivity check again shows no evidence of optimistic inflation from the transductive basis under this protocol. Component counts of 16, 32, 64, and 128 give 0.428, 0.434, 0.396, and 0.424, and the conference default of 64 gives the lowest of the four, with overlapping intervals. One control changes: over the long window, six bins beat the single-bin total count under 30% electrode removal (+0.032 BA, CI [+0.007, +0.056]) and lean the same way on clean data (+0.034, [−0.010, +0.076]), whereas on SHAPE’s short window BSC₆ and BSC₁ were indistinguishable. This is consistent with a window that spans effects at different latencies, here the salience effect followed by the valence effect, although the clean contrast alone does not exclude zero.

B. Feature-Coordinate Fills: What Replicates

Proposition 1 names the input-level cause of the conference result: the zero-filled block sits at $-\mu_S/\sigma_S$ in standardized coordinates. That cause replicates almost exactly. The mean $|\mu|/\sigma$ of the features is 1.20, 0.75, and 0.06 for band-power, ERP-window, and the reservoir on SHAPE, and 1.20, 0.72, and 0.06 on the external long epoch (1.20, 0.68, 0.05 at the matched epoch). The near-centeredness of the reservoir

TABLE I

CLEAN THREE-CLASS BALANCED ACCURACY (SUBJECT-LEVEL BOOTSTRAP 95% CI), MACRO-F1, MACRO ONE-VS-REST ROC-AUC FROM POOLED OUT-OF-FOLD PROBABILITIES, AND THE LARGEST BA AMONG 20 WITHIN-SUBJECT LABEL PERMUTATIONS. SHAPE VALUES USE THE CONFERENCE WINDOWS [2], AND THE RERUN OF THIS STUDY REPRODUCES THEM EXACTLY (TABLE II GIVES THE WINDOWS MEASURED FROM ONSET). THE CONFERENCE STORED NO EEGNET PROBABILITIES, SO F1 AND AUC ARE NOT AVAILABLE FOR IT. CHANCE IS 1/3 (AUC 0.5).

Cohort	Encoder	BA [95% CI]	F1	AUC	Null max
SHAPE ($N=211$)	ERP-window	0.629 [0.597, 0.660]	0.629	0.799	0.390
	Band-power	0.494 [0.460, 0.528]	0.492	0.682	0.378
	Reservoir	0.487 [0.447, 0.523]	0.485	0.658	0.379
	EEGNet	0.731 [0.701, 0.763]	–	–	–
	EEGNet + aug.	0.731 [0.700, 0.763]	–	–	–
External, $-200..800$ ms	ERP-window	0.386 [0.358, 0.414]	0.385	0.561	0.367
	Band-power	0.368 [0.339, 0.399]	0.367	0.531	0.365
	Reservoir	0.320 [0.288, 0.354]	0.320	0.482	0.368
	EEGNet	0.501 [0.471, 0.534]	0.501	0.666	–
	EEGNet + aug.	0.523 [0.491, 0.556]	0.522	0.679	–
External, long epoch	ERP-window	0.427 [0.398, 0.455]	0.427	0.597	0.361
	Band-power	0.389 [0.360, 0.418]	0.388	0.553	0.377
	Reservoir	0.396 [0.361, 0.428]	0.396	0.576	0.387
	EEGNet	0.447 [0.415, 0.478]	0.447	0.635	–
	EEGNet + aug.	0.488 [0.455, 0.523]	0.488	0.672	–

TABLE II

SHAPE WITH THE CONFERENCE WINDOWS AND WITH WINDOWS MEASURED FROM TRUE ONSET (BOTH FIXED BEFORE THEIR RESULTS WERE COMPUTED): CLEAN BA OF ERP-WINDOW AND THE RESERVOIR (BAND-POWER READS THE WHOLE EPOCH AND STAYS AT 0.494), THE PAIRED ERP-WINDOW MINUS RESERVOIR DIFFERENCE AT 30% DROPOUT UNDER ZERO AND MEAN FILL, AND THE PAIRED DIFFERENCE BETWEEN $\rho = 0$ AND $\rho = 0.9$ AVERAGED OVER TEN RESERVOIR DRAWS (CLEAN, SAME SUBJECT RESAMPLES FOR EVERY DRAW).

Window (ms)	ERP clean BA	Res. clean BA	Zero fill ERP–Res., 30% dropout	Mean fill ERP–Res., 30% dropout	$\rho=0 - 0.9$ draw mean
Conference ($-160..+73$)	0.629	0.487	+0.049 [+0.003, +0.095]	+0.092 [+0.049, +0.134]	–
Onset ($+39..+273$)	0.645	0.548	+0.044 [+0.000, +0.090]	+0.037 [–0.003, +0.077]	–

belongs to its PCA step, not to its spikes: the raw spike-count code is about as far from centered as band-power at the matched epoch (1.14 versus 1.20) and farther on the long window (2.58), and the shared basis subtracts the pooled mean.

The accuracy consequence depends on the window and the cohort (Table III, Fig. 4). On SHAPE with the conference windows, train-mean fill exceeded zero-fill at 30% dropout by +0.055 BA for band-power and +0.042 for ERP-window, and zero-fill halved the ERP-window lead over the reservoir (the rerun reproduces these values exactly). With the ERP windows measured from true onset, the band-power contrast is unchanged, since band-power reads the whole epoch, but the ERP-window contrast vanishes (-0.001 , CI $[-0.030, +0.028]$), and the ERP-window lead over the reservoir at 30% dropout is +0.044 under zero-fill and +0.037 under mean fill (Table II). The conference halving therefore rested on the first ERP window, which lies before onset. On the external long epoch the same contrasts are -0.013 ($[-0.038, +0.012]$) and $+0.014$ ($[-0.012, +0.039]$), and no fill moves the ERP-window lead outside noise. What does replicate is the near-centered code’s stability under the fill choice: the reservoir’s two constant fills differ by at most 0.011 BA at every rate from 10 to 50% on both external epochs, with paired intervals 0.009 to 0.026 BA wide on the long epoch against 0.041 to 0.060 for the non-centered encoders. Section V-C shows why accuracy moved on one cohort and

TABLE III

CHANNEL DROPOUT AT 30% (FEATURE-COORDINATE EXPERIMENT): SUBJECT-LEVEL BA UNDER FOUR FILL RULES, AND THE PAIRED DIFFERENCE ERP-WINDOW MINUS RESERVOIR UNDER THE SAME SUBJECT RESAMPLING. EXT. LONG AND EXT. MATCHED: EXTERNAL COHORT, LONG AND CONFERENCE-MATCHED EPOCHS.

Cohort	Fill	Band	ERP	Res.	ERP–Res [95% CI]
SHAPE	Zero	0.417	0.536	0.487	+0.049 [+0.003, +0.095]
	Mean	0.472	0.578	0.487	+0.092 [+0.049, +0.134]
	kNN	0.483	0.610	0.507	+0.103 [+0.062, +0.147]
	Spatial	0.488	0.589	0.482	+0.107 [+0.063, +0.150]
Ext. long	Zero	0.398	0.393	0.392	+0.001 [–0.039, +0.042]
	Mean	0.385	0.406	0.390	+0.017 [–0.025, +0.059]
	kNN	0.380	0.412	0.396	+0.016 [–0.026, +0.060]
	Spatial	0.385	0.398	0.404	–0.006 [–0.048, +0.035]
Ext. matched	Zero	0.360	0.370	0.325	+0.046 [+0.006, +0.085]
	Mean	0.360	0.354	0.325	+0.030 [–0.012, +0.072]
	kNN	0.376	0.389	0.319	+0.070 [+0.025, +0.113]
	Spatial	0.377	0.383	0.320	+0.063 [+0.022, +0.104]

not on the other.

C. Moving the Origin: A Direct Test of Proposition 1

Proposition 1 makes two predictions that can be tested by moving the coordinate origin. Both hold on SHAPE, where every encoder has accuracy to lose. On the external cohort the

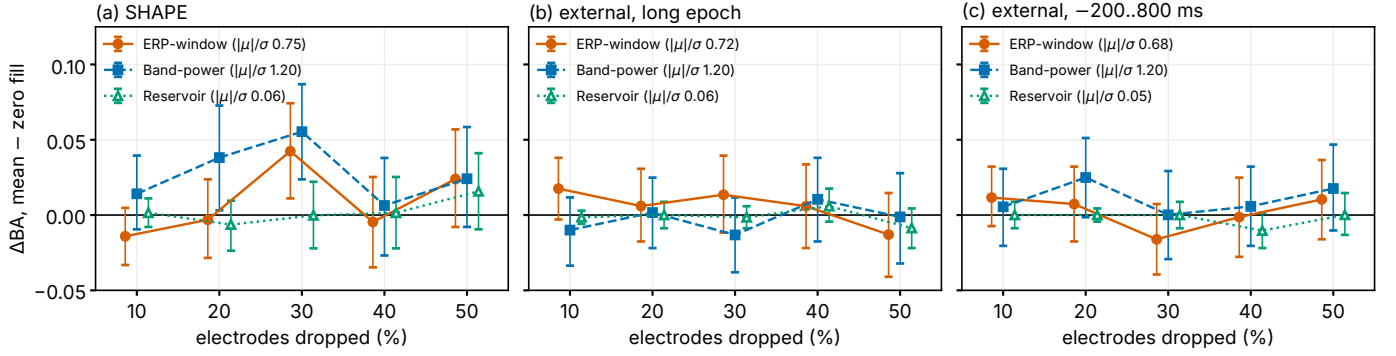


Fig. 4. How much the fill rule alone moves channel-dropout accuracy: paired subject-level change in BA (95% CI) of train-mean fill minus zero-fill against the share of electrodes dropped, for each fixed encoder. The legend gives the measured mean $|\mu|/\sigma$ of each encoder’s features on that cohort. (a) SHAPE (conference windows, reproduced exactly by the rerun). (b) External cohort, long epoch. (c) External cohort, conference-matched epoch. The input-level cause ($|\mu|/\sigma$) replicates on both cohorts, but the accuracy change does not, because the external encoders have little accuracy above chance to lose.

first holds on both epochs and the second on the long epoch (Fig. 5, Table IV).

First, clean predictions do not depend on the origin. Refitting every readout on features shifted by $b = \mu + 2\sigma$ changes no predicted label for band-power or the reservoir on either cohort. For ERP-window it changes at most 0.8% of the labels in any fold on SHAPE and 0.7% on the external long epoch, near-ties that the stopping tolerance of the L-BFGS solver resolves differently, and class probabilities differ by at most 5.1×10^{-3} and 5.8×10^{-3} (the reservoir’s agree to 1.1×10^{-13}).

Second, the dropout score depends on where the fill sits. On SHAPE, with 30% of the electrodes dropped, ERP-window BA is 0.578 with the fill at the training mean ($\kappa = 0$) but 0.392 and 0.381 with the same block three standard deviations above or below it, paired losses of 0.186 (CI [+0.148, +0.226]) and 0.198 (CI [+0.163, +0.234]), and its loss already excludes zero at $|\kappa| = 1$. Band-power falls from 0.472 to 0.409 and 0.392, again with losses at every $|\kappa| \geq 1$. The reservoir moves least. It loses 0.052 at $\kappa = +2$ (CI [+0.025, +0.081]) and 0.069 at $\kappa = +3$ (CI [+0.039, +0.100]), and no contrast at negative κ excludes zero. The conference’s zero-versus-mean comparison is one step along this axis. Each encoder’s native zero places its dropped features at a mean absolute standardized distance of 1.20 (band-power), 0.78 (ERP-window), and 0.06 (reservoir) from the training mean, and costs +0.055 (CI [+0.024, +0.087]), +0.042 (CI [+0.011, +0.074]), and -0.000 (CI [-0.022, +0.022]) BA against train-mean fill. The reservoir’s feature-coordinate zero (its PCA origin) is its training mean to within 0.06 standard deviations, which is why the fill choice was immaterial for it.

With the windows measured from onset the sweep keeps its form: ERP-window BA is 0.580 at $\kappa = 0$ and 0.439 and 0.438 at $\kappa = \pm 3$, and the reservoir loses 0.062 at $\kappa = +3$ (CI [+0.030, +0.095]). ERP-window’s native zero, however, now costs nothing against mean fill (-0.001, CI [-0.030, +0.028]) although it changes 21.6% of the labels. Proposition 1 predicts that a fill moves the input and therefore the labels. Whether the moved labels are more often wrong depends on where the shift sends observations relative to the decision boundaries, which a sweep over κ shows and a single zero-versus-mean comparison does not.

TABLE IV

ORIGIN TEST AT 30% DROPOUT (SHAPE): MEAN $|\mu|/\sigma$ OF THE DROPPED BLOCK, CLEAN BA, BA WITH THE NATIVE ZERO-FILL AND WITH THE FILL AT $\kappa = -2, 0, +2$, THE CLASS-CENTERED LOGIT SHIFT BETWEEN ZERO AND MEAN FILL, AND THE PERCENTAGE OF TEST LABELS THAT THE ZERO-VERSUS-MEAN CHOICE CHANGES.

Encoder	$\frac{ \mu }{\sigma}$	Clean	Zero	$\kappa=-2$	$\kappa=0$	$\kappa=+2$	$\ \Delta\ell\ $	Flip %
Band-power	1.20	0.494	0.417	0.411	0.472	0.412	4.13	40.9
ERP-window	0.78	0.629	0.536	0.411	0.578	0.412	1.75	27.7
Reservoir	0.06	0.487	0.487	0.477	0.487	0.434	0.43	7.6

The external long epoch shows the same form at smaller size. ERP-window BA is 0.406 at $\kappa = 0$ and 0.345 at $\kappa = +3$, a loss of 0.062 (CI [+0.034, +0.089]), and ERP-window also loses at $\kappa = \pm 2$. Band-power and the reservoir lose only at $\kappa = +3$ (0.036, CI [+0.001, +0.069], and 0.041, CI [+0.007, +0.073]). At the conference-matched epoch, where the encoders are near chance, no κ contrast at 30% dropout excludes zero.

The mechanism is the same on both cohorts even where accuracy cannot move. Between zero-fill and mean-fill, the class logits of a test observation move by $\|\Delta\ell\| = 4.13$ for band-power, 1.75 for ERP-window, and 0.43 for the reservoir on SHAPE (class-centered norm, mean over folds), and the fill choice changes the predicted label of 40.9%, 27.7%, and 7.6% of the test observations, in the order of the encoders’ non-centeredness. The external cohort gives the same order on the long epoch (34.5%, 32.9%, 3.6%) and at the matched epoch (37.8%, 25.9%, 3.3%). On SHAPE those label changes cost accuracy. On the external cohort they rewrite about a third of the non-centered encoders’ labels yet cost little, because the encoders have about a third of SHAPE’s above-chance margin to lose (0.32 for ERP-window, 0.34 for band-power). Proposition 1 thus predicts which representations a fill rule perturbs, and the above-chance margin limits how much accuracy that perturbation can remove.

D. The Empty Baseline

Proposition 2 predicts that under per-epoch z-scoring the pre-stimulus mean of each normalized trace is a scaled nega-

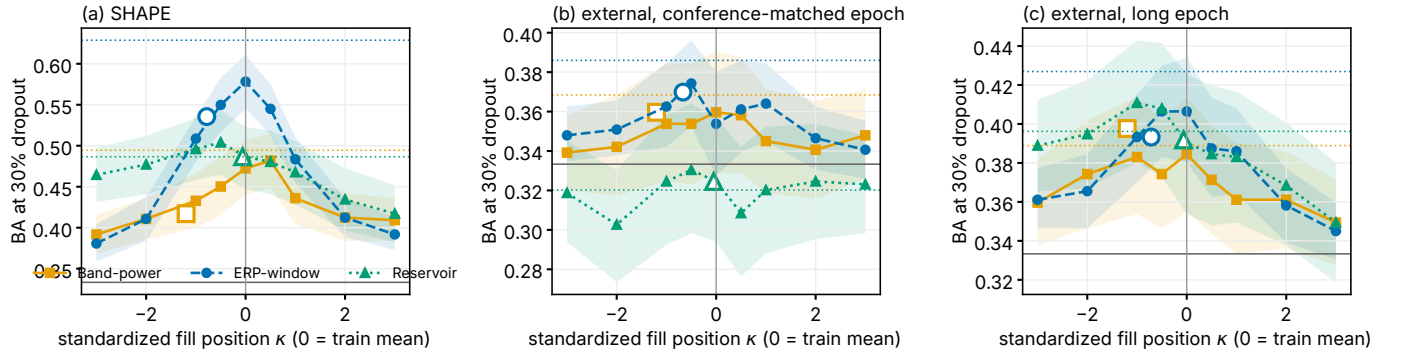


Fig. 5. Moving the fill along the standardized axis (30% of electrodes dropped). Subject-level BA (95% CI, shaded) versus the standardized position κ of the filled block ($\kappa = 0$: train-mean fill). Open markers: each encoder’s native zero-fill, plotted at minus the mean $|\mu|/\sigma$ of its dropped block for reference (the native zero places each feature at its own $-\mu_i/\sigma_i$, so it is not a single point on the κ axis). Dotted lines: clean BA, which does not depend on the origin (up to solver tolerance). (a) SHAPE: every encoder loses as the fill moves away from the training mean, ERP-window most and the reservoir least. (b) External cohort, conference-matched epoch, where every encoder is near chance. (c) External long epoch: the same form at smaller size.

tive of its post-stimulus mean. The correlation between the two is -0.92 across the 633×34 SHAPE traces (post-stimulus mean in microvolts), and -0.90 and -0.91 across the 684×32 external traces at the matched and long epochs, so the transfer happens on both cohorts. Whether it carries condition information depends on how much the post-stimulus mean differs between conditions (Fig. 6).

On SHAPE it carries a great deal. ERP-window features computed only from the 200 ms before onset reach BA 0.540 [$0.506, 0.575$] under per-epoch z-scoring (permutation $p = 0.005$, the smallest value 200 permutations can give, with 95th null percentile 0.368) and 0.330 [$0.292, 0.368$] under the fixed per-channel scale, a paired difference of $+0.210$ (CI [$+0.161, +0.258$]). Samples recorded before the picture appeared thus classify affect far above chance once each epoch is normalized by its own mean and standard deviation. The first window of the conference ERP-window encoder (-120 to 0 ms) lies inside that interval, and with the fixed scale the encoder falls from 0.629 to 0.545 (paired $+0.084$, CI [$+0.044, +0.123$]).

The conference reservoir read the same offset. With its conference window (-160 to $+73$ ms) it scores 0.487 under z-scoring but 0.308 [$0.272, 0.343$] under the fixed scale, which is chance. Its code reads the offset almost perfectly. Under z-scoring, the spike-bin code of a window wholly before onset predicts each trace’s post-stimulus mean with out-of-fold $R^2 = 0.999$ (0.993 for the conference window), and under the fixed scale with $R^2 = -0.093$ (0.012). The pre-specified per-channel correlation between the total spike count and the signed post-stimulus mean is weak (median -0.04 over channels, range -0.47 to $+0.82$), but the correlation with its magnitude is strong (median $+0.70$): offsets of either sign raise the count, and the count alone predicts the signed mean poorly ($R^2 = 0.003$), so the sign is carried by which units fire, not by how many. The external cohort gives the same picture ($R^2 = 0.999$ under z-scoring against -0.764 under the fixed scale for the pre-stimulus window). The conference accuracy of the reservoir on SHAPE therefore measures the normalization offset, a function of the post-stimulus mean, and not evoked dynamics inside its window. This is not a failure

of the reservoir under the fixed scale: with the same scale it reaches 0.580 after onset. This is consistent with the single-bin control BSC_1 matching BSC_6 there ($+0.001$ BA), since an offset that is constant within the window has no temporal structure to bin.

Pointed at the evoked signal, the same reservoir measures it. Over the whole post-onset interval (0 to $+797$ ms, same reservoir, same six bins) it reaches 0.615 [$0.581, 0.648$] under z-scoring, a gain of $+0.128$ BA over the conference window (CI [$+0.081, +0.174$]), and 0.580 [$0.545, 0.615$] under the fixed scale, where the difference between the normalizations no longer excludes zero ($+0.035$, CI [$-0.006, +0.074$]). Its point estimate is then close to that of the conference ERP-window encoder (0.629). The same normalization removes information where it belongs, although on SHAPE the effect is not resolved: post-onset ERP-window features reach 0.564 under the fixed scale and 0.547 under z-scoring (paired -0.017 , CI [$-0.054, +0.019$]).

The external cohort replicates the direction at smaller size. Pre-stimulus ERP-window features reach 0.357 [$0.333, 0.382$] under z-scoring at the matched epoch (permutation $p = 0.075$) and 0.367 [$0.342, 0.392$] on the long epoch ($p = 0.025$), against 0.325 and 0.325 under the fixed scale, although the long-epoch paired difference does not exclude zero ($+0.042$, CI [$-0.001, +0.085$]). The conference reservoir window is at chance at the matched epoch (0.320), and moved after onset it gains $+0.092$ BA (CI [$+0.045, +0.139$]), reaching 0.412 , a point estimate above that of the ERP-window encoder (0.386). On the long epoch, post-onset ERP-window features reach 0.447 under the fixed scale and 0.406 under z-scoring (paired -0.041 , CI [$-0.075, -0.007$]), because dividing each epoch by its own standard deviation discards amplitude differences that carry the effect.

E. The Edge of Stability Across Reservoir Draws

The conference located the transition of draw 42 at $\rho^* \approx 0.82$ on SHAPE drives. On the external drives the same draw gives $\rho^* = 0.82$, and its order-parameter curve agrees with the SHAPE curve within 0.004 at every grid point from $\rho = 0$ to 2 (the ten-draw mean in Fig. 7a saturates lower), consistent

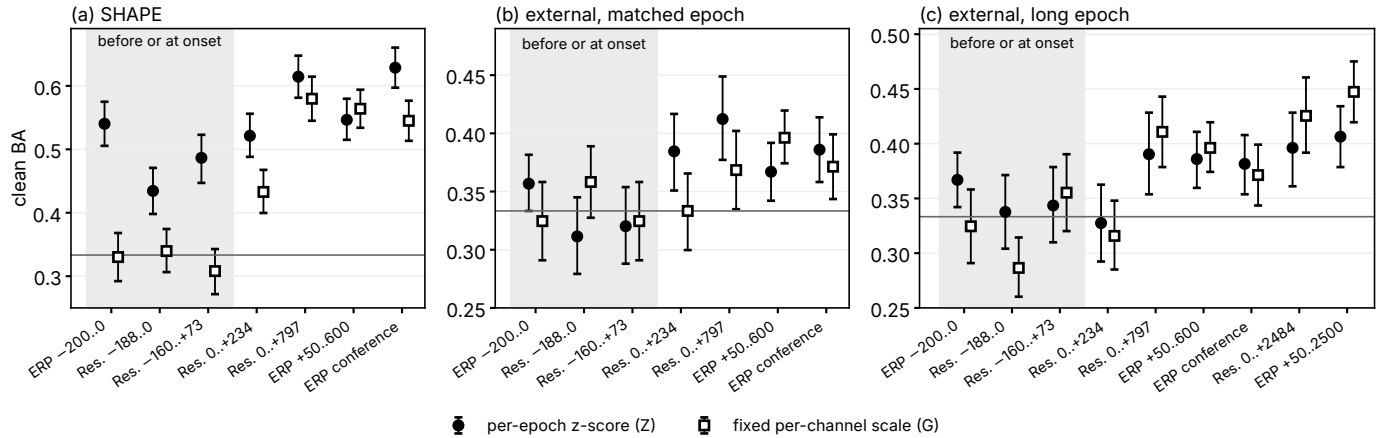


Fig. 6. What each window can read. Subject-level BA (95% CI) under per-epoch z-scoring (filled) and a fixed per-channel scale (open), for windows before or at onset (shaded) and after onset. (a) SHAPE: under z-scoring, pre-stimulus ERP-window features and the conference reservoir window are far above chance, and under the fixed scale both fall to chance, whereas the post-onset reservoir window keeps its accuracy. (b) External cohort, conference-matched epoch. (c) External long epoch, with two added windows after onset (0 to +2484 ms for the reservoir, the four post-onset ERP windows). Chance = $1/3$.

with a transition set by the reservoir and the statistics of its z-scored drive rather than by the cohort. Across ten draws, $\rho^* = 0.76 \pm 0.04$ (range 0.69 to 0.84, Table V). Every draw places the transition below the echo-state heuristic $\rho = 1$, and the conference draw lies near the top of the range. The unit moves the transition more than the draw does. At thresholds 0.25, 0.5, and 1.0 the transition lies at 0.35, 0.82, and 1.57 (Fig. 7c, where at $\vartheta = 1.0$ the damage has not saturated by $\rho = 2$, so this value is a lower bound), and from leak 0.02 to 0.2 it rises from 0.79 to 0.95. The heuristic $\rho = 1$ therefore marks no transition for this spiking unit: depending on the threshold, the measured edge lies well below it or above it.

Accuracy on the long epoch follows the conference pattern in shape but not in strength (Fig. 7b, Table V). Averaged over five draws, clean BA is lowest without recurrence ($\rho = 0$, 0.372) and in the strongly irregular regime ($\rho = 1.5$, 0.373), and highest at $\rho = 1.2$ (0.413), the most accurate setting in four of five draws, against 0.385 at the conference operating point. No single draw, however, separates $\rho = 0$ or $\rho = 1.5$ from $\rho = 0.9$ with a subject-level interval that excludes zero, so the recurrence effect measured on SHAPE (-0.049 BA, CI $[-0.098, -0.002]$) is not established on the external cohort, whose margin is smaller. The optimum is broad, spans the measured transition, and does not track ρ^* draw by draw.

Electrode removal separates the regimes more sharply than clean accuracy does. With 30% of the electrodes silent, mean BA over the five draws is 0.386 and 0.389 at $\rho = 0.3$ and 0.6 but 0.339, 0.339, and 0.340 at $\rho = 0.9$, 1.2, and 1.5, close to chance on average (single draws stay above chance in some cases). The loss to removal is largest past the transition (0.012 BA at $\rho = 0.6$, 0.045 at 0.9, 0.074 at 1.2, and 0.033 at 1.5), as is the firing rate, which rises with ρ (0.05, 0.14, and 0.23 spikes per unit and step). A silent electrode is a fixed point of every reservoir, so its block is zero spikes whatever ρ is. Once activity rises past the transition, that silence lies far from the reservoir's typical state. For draw 42, in standardized readout coordinates the silent-electrode block ($-P\bar{b}$ after PCA) sits on average 1.21 to 1.36 standard deviations from the training

mean for $\rho \leq 0.6$, but 4.95 at $\rho = 0.9$ and 6.23 at $\rho = 1.2$, and the raw spike counts move from $|\mu|/\sigma = 0.90$ to 4.40. Past the measured edge of stability, nothing, in the literal sense of a silent electrode, lies far from anything the readout has seen. SHAPE, read through the short conference window, showed no such separation: the loss to 30% removal was 0.013, 0.014, and 0.014 BA at $\rho = 0.9$, 1.2, and 1.5.

F. DEAP Check

On DEAP [47] (conference result, Fig. 8a), band-power features scored 0.540 under zero-fill at 30% dropout but 0.621 under train-mean and 0.638 under k NN fill, against 0.639 clean, so the fill rule alone moved dropout accuracy by up to 0.11 BA on a second dataset, task, and montage. Band-power had a margin of 0.14 above the binary chance level there, enough for the fill-induced label changes to cost accuracy, as on SHAPE.

G. Signal-Level Perturbations, Physical Repair, and Recorded Noise

At the signal level every encoder sees the same corrupted test signal (Tables VI and VII, Fig. 9). On SHAPE (conference results), ERP-window features led the reservoir in every condition, the reservoir held its accuracy through 30% electrode removal but fell to 0.393 at 50%, 5 dB noise cost band-power 0.059 BA, and ± 50 ms jitter moved no fixed encoder.

The chance floor in practice. At the external conference-matched epoch, where no fixed encoder is far from chance, perturbations move BA in both directions: the reservoir scores 0.320 clean and 0.370 under 5 dB noise, a raw retention of 1.16. A ratio above one here measures nothing about stability. It is the raw-retention artifact of the conference paper appearing without being sought. We therefore read the external signal-level results on the long epoch.

Electrode removal and spline repair. On the long epoch, removing electrodes as zero traces costs the reservoir 0.038 BA at 30% (CI $[+0.004, +0.073]$) and 0.051 at 50%

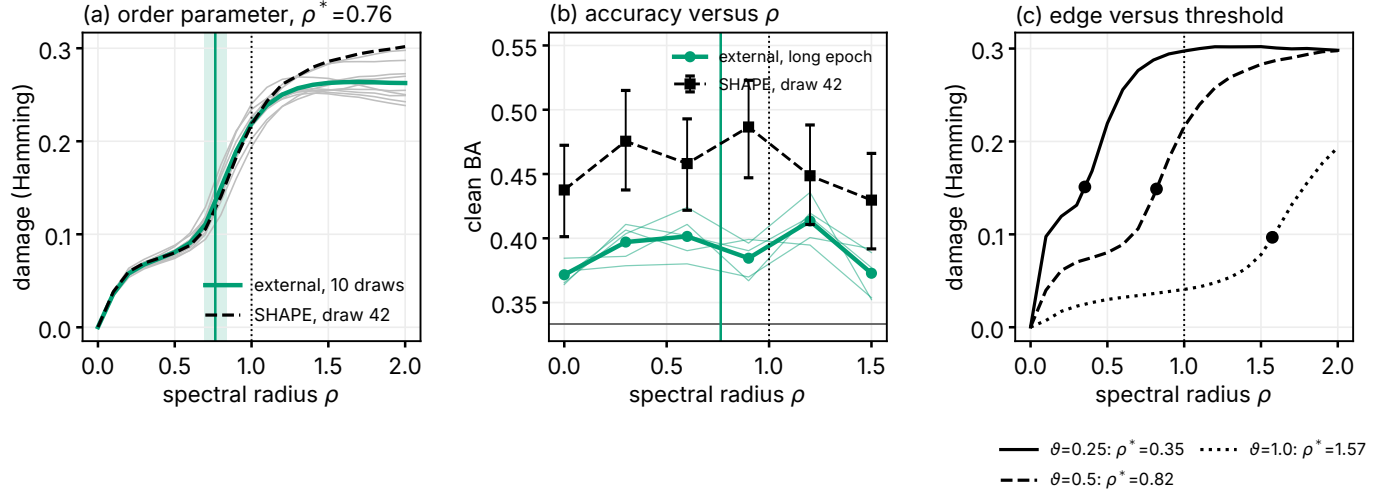


Fig. 7. The measured edge of stability. (a) Damage-spreading order parameter versus spectral radius for ten reservoir draws on the external drives (thin), their mean (thick), and SHAPE (draw 42, dashed). Shading spans the transitions ρ^* of the ten draws, the solid line marks their mean, and the dotted line the echo-state heuristic $\rho = 1$. (b) Clean BA versus ρ for five draws on the external long epoch (thin) and their mean (thick), with SHAPE (draw 42, 95% CI). $\rho = 0$ removes the recurrence, and chance = $1/3$. (c) Order parameter at three firing thresholds (draw 42), with dots at each ρ^* .

TABLE V
EDGE OF STABILITY ACROSS RESERVOIR DRAWS (EXTERNAL COHORT): MEASURED TRANSITION ρ^* (ORDER PARAMETER ON THE CONFERENCE-MATCHED DRIVES) AND CLEAN BA ON THE LONG EPOCH AT EACH SPECTRAL RADIUS, THE MOST ACCURATE ρ , AND THE PAIRED DIFFERENCE BETWEEN $\rho = 0$ AND $\rho = 0.9$ UNDER THE SAME SUBJECT RESAMPLING.

Draw	ρ^*	0	0.3	0.6	0.9	1.2	1.5	Best	$\rho=0 - 0.9$ [95% CI]
42	0.82	0.365	0.404	0.424	0.396	0.436	0.352	1.2	-0.031 [-0.077, +0.016]
43	0.69	0.364	0.411	0.402	0.390	0.417	0.377	1.2	-0.026 [-0.073, +0.019]
44	0.76	0.374	0.379	0.380	0.370	0.401	0.392	1.2	+0.004 [-0.047, +0.054]
45	0.76	0.370	0.406	0.390	0.399	0.395	0.354	0.3	-0.029 [-0.072, +0.016]
46	0.78	0.385	0.386	0.411	0.367	0.420	0.389	1.2	+0.017 [-0.029, +0.064]
Mean	0.76	0.372	0.397	0.401	0.385	0.413	0.373		
30% silent		0.357	0.386	0.389	0.339	0.339	0.340		

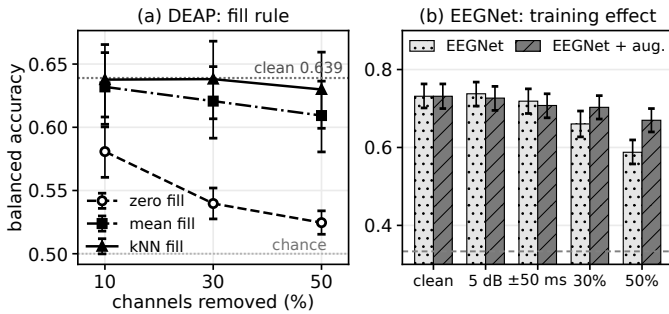


Fig. 8. Conference results [2]. (a) DEAP [47] (32 subjects, within-subject binary valence, band-power, CIs from resampling subjects): under channel removal, BA depends strongly on the fill rule. (b) SHAPE, EEGNet without and with channel-dropout and noise augmentation under the signal-level conditions of Table VI (subject-level 95% CIs): augmentation leaves clean accuracy unchanged and cuts the loss to electrode removal by more than half.

($+0.015, +0.088$), and band-power 0.043 at 50%, while ERP-window features lose 0.011 at 50%. ERP-window therefore leads the reservoir by $+0.070$ (CI $[+0.041, +0.101]$) with half the electrodes removed, a larger lead than on clean data. Repairing the same electrodes by spherical-spline interpolation (correlation 0.84 with the true trace at 50%) removes that lead: with the repaired signal no fixed encoder loses more

than 0.011 BA at 50%, the reservoir gains $+0.062$ over the zero trace (CI $[+0.025, +0.101]$), band-power gains $+0.050$ ($[+0.018, +0.082]$), and the ERP-window lead falls to $+0.010$ ($[-0.032, +0.051]$). The signal-level ranking under electrode loss was set by representing the absent electrode as a zero trace. A zero trace is a silent drive, and for the reservoir silence lies far from its typical state (Sec. V-E).

Noise and jitter. The SHAPE pattern replicates: 5 dB noise costs band-power 0.047 BA (CI $[+0.015, +0.079]$, against 0.059 on SHAPE) and leaves ERP-window and the reservoir within noise, and ± 50 ms jitter moves no fixed encoder.

Recorded observation noise. Re-averaging each test observation from fewer of its own trials lowers ERP-window BA from 0.427 to 0.367 at one trial (loss 0.060, CI $[+0.018, +0.102]$), whereas the reservoir is unchanged at one trial (0.396, loss 0.000, CI $[-0.045, +0.045]$) and band-power loses 0.023 ($[-0.019, +0.064]$) (Fig. 9a). The paired ERP-window minus reservoir difference moves from $+0.030$ on clean data to -0.029 at one trial (CI $[-0.072, +0.015]$), so the encoders are not separable on single trials. Training used the full averages throughout, so these numbers measure how each fixed representation transfers from trial averages to noisier observations.

Training effect. EEGNet, trained end to end on the signal, is

TABLE VI

SHAPE, SIGNAL-LEVEL PERTURBATIONS (CONFERENCE RESULTS [2]): SUBJECT-LEVEL BA WITH EVERY REPRESENTATION RECOMPUTED FROM THE CORRUPTED TEST SIGNAL. PER-CELL 95% CIs ARE ± 0.03 TO ± 0.04 .

Condition	Band	ERP	Res.	EEGNet	+aug.
Clean	0.494	0.629	0.487	0.731	0.731
10% removed	0.476	0.602	0.469	0.719	0.722
30% removed	0.417	0.536	0.474	0.660	0.703
50% removed	0.450	0.474	0.393	0.588	0.670
Noise 5 dB	0.436	0.619	0.479	0.738	0.727
Jitter ± 50 ms	0.496	0.626	0.493	0.719	0.708

the most accurate encoder at the external cohort’s conference-matched epoch, as on SHAPE. There, where every fixed encoder is near chance, it reaches 0.501 (CI [0.471, 0.534]) without and 0.523 with augmentation, +0.116 above ERP-window (CI [+0.076, +0.154]), and its paired lead over ERP-window has a lower bound at or above zero in every condition (Table VIII). The SHAPE training effect does not replicate in size. On SHAPE, channel-dropout and noise augmentation cut EEGNet’s loss to electrode removal by more than half (0.144 to 0.062 BA at 50%). Here, over all five partitions with the default recipe, augmentation adds at most +0.041 BA in any condition, with intervals that exclude zero only under 10% removal, 30% spline repair, and 5 dB noise (+0.041, CI [+0.010, +0.072]), and the loss is 0.058 and 0.050 at 30 and 50% without augmentation and 0.051 and 0.057 with it. Across 4 training recipes on the seed-42 partition (Table X), augmentation raised BA at 50% removal by +0.020 to +0.063 (intervals excluding zero for 3 of 4 recipes) but also raised clean BA by +0.013 to +0.046 (point estimates, intervals excluding zero for 1 of 4), so it reduced the loss to removal by only 0.007 to 0.025 BA, never by half. The seed-42 partition favors augmentation: with the default recipe its gain at 50% removal is +0.047 (CI [+0.013, +0.079]) there, against +0.015 (CI [−0.013, +0.042]) over five partitions. Spline repair recovers most of the removal loss for EEGNet as for the fixed encoders (loss 0.009 at 50%, CI [−0.018, +0.037]). The trained network is also the encoder most affected by recorded observation noise: re-averaging test observations from one trial costs it 0.131 BA (CI [+0.092, +0.171]), against 0.055 for ERP-window, consistent with a network fitted to trial averages. On the long epoch (seed-42 partition), EEGNet reaches 0.447 without and 0.488 with augmentation, against 0.463 and 0.485 at the matched epoch on the same partition, so with this recipe the late effects that help the fixed encoders do not raise its accuracy, and its clean lead over ERP-window is +0.036 (CI [−0.003, +0.076], +0.017 at 50% removal, CI [−0.020, +0.056]). Augmentation adds accuracy with intervals excluding zero only on clean data (+0.041, CI [+0.010, +0.073]) and with 50% spline repair (+0.038, CI [+0.007, +0.069]), and without reducing the loss to 50% removal (0.053 without and 0.064 with augmentation), and spline repair again recovers most of that loss (0.013).

VI. DISCUSSION

Absence is a coordinate. The conference result was that zero-fill halves an encoder difference. The rerun shows what

TABLE VII

EXTERNAL COHORT, LONG EPOCH: SUBJECT-LEVEL BA WITH EVERY REPRESENTATION RECOMPUTED FROM THE CORRUPTED TEST SIGNAL (SAME DRAWS FOR ALL ENCODERS). SPLINE: THE REMOVED ELECTRODES REPAIRED BY SPHERICAL-SPLINE INTERPOLATION. n TRIALS: EACH TEST OBSERVATION RE-AVERAGED FROM n OF ITS OWN TRIALS. EEGNET ON THIS EPOCH USES THE SEED-42 PARTITION ONLY. PER-CELL 95% CIs ARE ± 0.02 TO ± 0.04 .

Condition	Band	ERP	Res.	EEGNet	+aug.
Clean	0.389	0.427	0.396	0.447	0.488
10% removed	0.386	0.404	0.367	0.444	0.468
30% removed	0.398	0.393	0.358	0.412	0.439
50% removed	0.346	0.415	0.345	0.395	0.424
10% spline	0.376	0.405	0.404	–	–
30% spline	0.386	0.399	0.393	0.453	0.484
50% spline	0.396	0.417	0.406	0.434	0.472
Noise 5 dB	0.342	0.420	0.405	0.462	0.484
Jitter ± 50 ms	0.389	0.418	0.421	0.447	0.472
1 trial	0.367	0.367	0.396	0.363	0.358
4 trials	0.349	0.406	0.377	0.392	0.411
16 trials	0.368	0.399	0.393	0.466	0.472

that rested on and what generalizes. What generalizes is the input-level geometry and its consequence for the dropout score. On both cohorts each encoder’s dropped block sits at nearly the same standardized distance from its training mean, the fill choice rewrites labels in the order of that distance, and moving the fill along the standardized axis changes the dropout score while clean predictions stay fixed up to solver tolerance, by up to 0.198 BA for SHAPE’s ERP-window features. What does not generalize is any single zero-versus-mean contrast. Its size depends on where the native zero falls relative to the decision boundaries, which changed with the ERP windows (the conference halving vanished when they were measured from onset), and on the above-chance margin, which limits what a fill can cost (the external cohort). A dropout comparison that zero-fills is therefore a comparison of coordinate systems as much as of representations. This agrees with the observations that channel-dropout fragility of EEG foundation models depends on whether dropped channels are removed or zero-padded [33] and that how a channel goes missing or out of place changes measured robustness [1], and Proposition 1 with the origin sweep gives the mechanism and its size for standardized features. The reservoir’s indifference to the fill belongs to its PCA coordinates: its raw spike counts are about as far from centered as band-power, and moved far from its own mean its code degrades too.

The empty baseline is not empty after normalization. Per-epoch z-scoring is a common default, and Proposition 2 shows that it writes $-\frac{|Q|}{T}m_Q/s$ into every pre-stimulus sample. On SHAPE that is enough for the 200 ms recorded before the picture to classify affect at 0.540 BA, and for the conference reservoir, whose window lies mostly before onset, to reach its conference accuracy from the offset: under a fixed scale it falls to chance, and its code predicts the post-stimulus mean almost perfectly. The mechanism resembles the way high-pass filters move late ERP effects to earlier latencies [37]–[39], and, because the normalization uses the whole epoch, it is a form

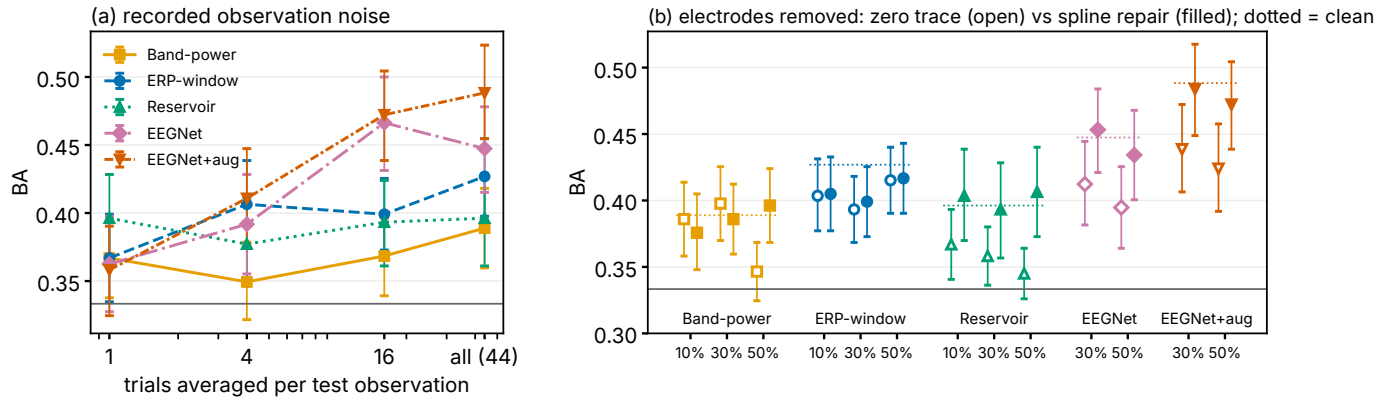


Fig. 9. Signal-level perturbations on the external cohort (long epoch). (a) Recorded observation noise: subject-level BA (95% CI) when each test observation is re-averaged from 1, 4, or 16 of its single trials, and with all trials. (b) Electrodes removed as a zero trace (open) or repaired by spherical-spline interpolation (filled), with dotted lines at clean BA. Chance = 1/3.

of leakage from the post-stimulus signal into the baseline [41]. It moves information within a subject’s own epoch, not across subjects, so it does not inflate accuracy the way cross-subject leakage does, but it misattributes it: a window read as early dynamics measures late amplitude. The conference accuracy of the reservoir, and the recurrence effect measured through the same window (Sec. V-E), therefore describe how the reservoir encodes a normalization offset, not evoked dynamics. Placed after onset, the same reservoir measures the evoked signal on both cohorts, close to the ERP-window encoder on SHAPE, and its temporal bins then carry information that the conference window’s did not. Two practical checks follow: decode from the pre-stimulus interval alone, and repeat the analysis with a fixed scale.

The edge of stability. The conference reported a transition near $\rho^* \approx 0.82$ for one reservoir draw. Ten draws on a second cohort’s drives place it at 0.76 ± 0.04 , always below $\rho = 1$, but the firing threshold moves it from 0.35 to at least 1.57. For a thresholded spiking reservoir the edge is therefore a measured property of the unit and its drive rather than a value fixed by the spectral radius, in line with edge-of-chaos analyses of other reservoirs [25]–[27] and with the result that $\rho < 1$ is neither necessary nor sufficient for the echo-state property [28]. Clean accuracy was broad around the transition on both cohorts. Its sharpest dependence on the operating point appeared under electrode removal: past the transition a silent electrode lies far from the reservoir’s typical code, and on the external long epoch accuracy with 30% of the electrodes silent fell close to chance on average, although SHAPE’s short window showed no such separation. An operating point chosen near the edge for clean accuracy may thus be the one least tolerant of absent inputs, and measuring both is cheap.

Training and physical repair. Two conference conclusions about electrode loss change with the second cohort. First, augmentation reduced EEGNet’s loss to electrode removal on the external cohort by at most 0.025 BA across four recipes on one partition, and not at 50% removal over five partitions with the default recipe. The conference finding that the training procedure sets this architecture’s accuracy under electrode loss therefore holds for SHAPE, where augmentation halved the

loss, and not as a general property of the architecture. On the external cohort augmentation acted mostly as a general gain in accuracy. Second, physical repair changed the ranking that the zero trace produced. With spline-repaired electrodes no fixed encoder lost more than 0.011 BA at 50% removal on the long epoch, EEGNet recovered most of its removal loss, and the ERP-window lead over the reservoir disappeared, although at the conference-matched epoch, where the encoders are near chance, the lead persisted after repair. A zero trace is itself a fill, the signal-level counterpart of zero-fill, and the same advice applies: state it, and compare it with a physically motivated repair [34] before ranking representations by their tolerance of missing electrodes.

A reporting standard. The results support five reporting items for EEG representation studies under perturbation. First, absolute accuracy with subject-level uncertainty and an empirical permutation floor [42], [43], never retention ratios alone. Second, the fill rule, varied along the standardized axis rather than as one contrast, together with the mean $|\mu|/\sigma$ of the features and the above-chance margin, which together limit how much the rule can matter. Third, the normalization, whether any window used for decoding precedes onset, and the accuracy of the pre-stimulus interval alone. Fourth, trained-model results with and without augmentation [32]. Fifth, for reservoirs, the measured transition ρ^* and more than one weight draw.

Limitations. The SHAPE export documents neither its channel order nor its single trials, so spline repair and recorded trial-count noise were measured on the external cohort only, whose above-chance margin is about a third of SHAPE’s. The windows measured from onset were fixed in advance, but they are two of many possible windows, and no others were analyzed. The external analysis used one pre-specified long epoch beyond the conference-matched one. The reservoir’s draws, thresholds, and leaks were varied only around the conference design, EEGNet is the only trained architecture and was not rerun on SHAPE (it reads the whole epoch, so the window question does not arise for it), and DEAP was not reanalyzed. The fixed per-channel scale is pooled over all observations without labels, which is transductive with respect

to subject inclusion. The external cohort's authors interpolated some bad channels before release, which may favor spline repair there. Clinical labels, graph topology, and structure-function coupling are outside the scope of this paper.

VII. CONCLUSION

What an evaluation lets stand for nothing is part of the measurement. A zero-filled electrode sits at the coordinate origin, so its effect on dropout accuracy is set by how far that origin lies from the data and by how much the encoder has above chance. Per-epoch normalization writes the evoked signal into the baseline, so the empty interval before onset can carry condition information while the evoked signal itself loses some. The edge of stability of a fixed spiking reservoir is a measured quantity that moves with the weight draw and, more strongly, with the firing threshold, so the echo-state heuristic $\rho = 1$ marks no transition for this unit, and once activity rises past the transition, silence lies far from the reservoir's typical state. On two cohorts the reservoir is not the most accurate encoder, but its near-centered code and its controllable dynamics make each of these effects visible, and with its window placed after onset it measures the evoked signal it was built to measure.

ETHICS

The SHAPE study was collected by the Stony Brook University Laboratory for Clinical Affective Neuroscience under that laboratory's Institutional Review Board oversight. The external cohort and DEAP are publicly released datasets governed by their original collection protocols [12], [46], [47].

DATA AVAILABILITY

For the SHAPE dataset, contact the Laboratory for Clinical Affective Neuroscience at Stony Brook University [45]. The external cohort is public under a CC BY-NC 4.0 licence [46], and DEAP is public [47]. The analysis code, which produces every number, table, and figure from these data, and the aggregate results are available from the authors.

APPENDIX A SUPPLEMENTARY TABLES

Table VIII gives the signal-level results of the external cohort at the conference-matched epoch, where every fixed encoder is near chance, and Table IX the origin test on both external epochs. Table X reports EEGNet across training recipes on the seed-42 partition of the conference-matched epoch.

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TABLE VIII
EXTERNAL COHORT, CONFERENCE-MATCHED EPOCH (−200 TO +800 MS): SUBJECT-LEVEL BA UNDER THE SIGNAL-LEVEL CONDITIONS OF TABLE VII.

Condition	Band	ERP	Res.	EEGNet	+aug.
Clean	0.368	0.386	0.320	0.501	0.523
10% removed	0.352	0.379	0.300	0.465	0.497
30% removed	0.360	0.370	0.310	0.443	0.472
50% removed	0.342	0.349	0.301	0.452	0.466
10% spline	0.367	0.392	0.326	—	—
30% spline	0.361	0.374	0.351	0.487	0.516
50% spline	0.357	0.401	0.346	0.493	0.497
Noise 5 dB	0.354	0.392	0.370	0.459	0.500
Jitter ±50 ms	0.365	0.404	0.355	0.477	0.480
1 trial	0.329	0.330	0.357	0.370	0.377
4 trials	0.358	0.383	0.344	0.424	0.431
16 trials	0.361	0.392	0.346	0.484	0.482

TABLE IX
ORIGIN TEST AT 30% DROPOUT, EXTERNAL COHORT (COLUMNS AS IN TABLE IV). TOP: LONG EPOCH. BOTTOM: CONFERENCE-MATCHED EPOCH.

Encoder	$\frac{ \mu }{\sigma}$	Clean	Zero	$\kappa=-2$	$\kappa=0$	$\kappa=+2$	$\ \Delta\ell\ $	Flip %
Band-power	1.20	0.389	0.398	0.374	0.385	0.361	1.76	34.5
ERP-window	0.72	0.427	0.393	0.365	0.406	0.358	1.45	32.9
Reservoir	0.06	0.396	0.392	0.395	0.390	0.368	0.17	3.6

Encoder	$\frac{ \mu }{\sigma}$	Clean	Zero	$\kappa=-2$	$\kappa=0$	$\kappa=+2$	$\ \Delta\ell\ $	Flip %
Band-power	1.20	0.368	0.360	0.342	0.360	0.341	1.78	37.8
ERP-window	0.67	0.386	0.370	0.351	0.354	0.346	0.71	25.9
Reservoir	0.05	0.320	0.325	0.303	0.325	0.325	0.17	3.3

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TABLE X

EEGNET ACROSS TRAINING RECIPES (EXTERNAL COHORT, CONFERENCE-MATCHED EPOCH, SEED-42 PARTITION): SUBJECT-LEVEL BA CLEAN, WITH 30% AND 50% OF ELECTRODES REMOVED, AND UNDER 5 dB NOISE, WITHOUT AND WITH AUGMENTATION, AND THE PAIRED AUGMENTATION EFFECT AT 50% REMOVAL.

Recipe	EEGNet				EEGNet + aug.				Aug.—unaug., 50% [95% CI]
	Clean	30%	50%	Noise	Clean	30%	50%	Noise	
Default (80 ep., $p=0.25$)	0.463	0.408	0.399	0.434	0.485	0.447	0.446	0.465	+0.047 [+0.013, +0.079]
40 epochs	0.459	0.417	0.404	0.430	0.472	0.442	0.424	0.468	+0.020 [−0.009, +0.048]
160 epochs	0.431	0.408	0.377	0.421	0.477	0.437	0.440	0.484	+0.063 [+0.026, +0.096]
Dropout 0.5	0.463	0.409	0.411	0.459	0.485	0.456	0.442	0.477	+0.031 [+0.001, +0.060]

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